4.10 Mix Selection

An HMA pavement surface must serve the demands of multiple traffic loadings and environments. The HMA mixture chosen for the project must meet the project requirements. The pavement designer has more choices now than in the past for the type of mix selected and for the materials (aggregates and asphalt binders) to be used on the project. This has led to the development of specialized mixes to address very specific problems. New design procedures such as the Superpave design procedure and others discussed earlier in this chapter have been implemented. In many cases, higher-quality materials are being specified than are needed for the particular project. Because of these concerns, the National Asphalt Pavement Association—in a coordinated effort with the Federal Highway Administration (FHWA) and an Expert Task Group consisting of representatives from the paving industry and Departments of Transportation—developed National Asphalt Pavement Association Information Series 128, *HMA Pavement Mix Type Selection Guide.* This document provides detailed information on mix selection. As described earlier in this chapter, there are three primary types of mixes used for the construction of HMA pavements: fine and coarse graded dense mixes, stone matrix asphalt (SMA) mixes, and Open-Graded Friction Courses (OGFC) mixes.

4.10.1 Pavement Layers and Traffic Levels (definitions)

The following provides a short description of each of these layers in a pavement structure. Generally, materials of lesser quality are needed for each layer as they progress down in the pavement structure.

- Surface layer—contains the highest-quality materials. It provides friction, smoothness, noise control, rutting resistance and durability.
- Intermediate layer or binder layer—consists of one or more lifts of HMA below the surface layer. It provides structural strength to the pavement.
- Base layer—consists of one or more lifts of HMA placed at the bottom of the pavement structure.

The mix selected for use in an HMA pavement will vary depending on the traffic level of the pavement. Table 4.12 provides some definitions and the typical roadway application represented by that traffic level.

Table 4.12
Traffic Level Definitions

Traffic Designation	Equivalent 18,000 lb Single Axle Loads (ESALs) (20 yr design life)	Typical Roadway Applications
Low	< 300,000	• Roadways with very light traffic volumes such as local roads, county roads, and city streets, where truck traffic is prohibited or at a minimum. • Traffic considered local in nature, not regional, intra-state, or interstate. • Special-purpose roadways serving recreational sites or areas.
Moderate	300,000 to < 10,000,000	• Two-lane, multilane, divided, and partially or completely controlled access roadways. • Medium to highly trafficked city streets, state routes, U.S. highways and some rural interstates.
High	> 10,000,000	• Two-lane, multilane, divided and partially or completely controlled access highways. • Medium to highly trafficked city streets, state routes, U.S. highways and many interstate highways. • Truck-weighing stations or truck-climbing lanes on two-lane roadways.

4.10.2 Dense Graded Mixes

A dense graded mix is a mixture where the aggregate particles have an even distribution from coarse to fine. The goal is that the mix chosen will be stable and durable. To achieve these, it is generally necessary to introduce enough asphalt into the mix to insure durability but not so much as to reduce the stability of the mix below that needed for the traffic levels on the pavement. To insure that the chosen mix performs properly, it also must be built as it is designed and proper quality control procedures must be implemented. A special concern is the need for obtaining proper compaction on the roadway.

Dense graded mixes are defined by their nominal maximum aggregate size (defined earlier in this chapter and in Chapter 3, "Mineral Aggregates and Mixture Additives") and can be further defined as either fine-graded or coarse-graded. A fine-graded mix will have a higher proportion of fine sand-sized particles than a coarse-graded mix. Table 4.13 presents the criteria for fine/coarse graded HMA mixes. Coarse mixes are defined as having a combined aggregate gradation that passes below the primary control sieve control point, and fine mixes are defined as having a gradation that passes above or through the primary control sieve control point.

In general, the fine gradation mixes with the smaller nominal maximum aggregate sizes will be used for the surface mixes (top 1½ to 2 inches).

Table 4.13
Coarse Versus Fine Graded Mixes

Nominal Maximum Aggregate Size	Control Sieve	Breakpoint
9.5 mm	2.36	47%
12.5 mm	2.36	39%
19 mm	4.75	47%
25 mm	4.75	40%
37.5 mm	9.5	47%

These mixes will have higher asphalt contents and lower permeability (due to greater aggregate surface area and smaller size voids), which will result in improved durability (less block and shrinkage cracking). For applications where the surface will receive little or no vehicle traffic, such as bike trails and walking paths, consideration should be given to designing the mix with 2½ percent air-voids rather than 4 percent air-voids (standard requirement) to increase the durability of the mix further and reduce non-load associated cracking.

The lower levels (binder and base courses) should use the larger aggregate sizes. These mixes provide increased resistance to pavement deformation. This allows the placement of thicker lifts and requires less asphalt cement due to less surface area.

The lift thickness for all mixes (surface, binder and base) is dictated by the nominal maximum aggregate size. If the nominal maximum aggregate size is too large with relation to the lift thickness it will be very difficult to obtain proper compaction of the mat. It is recommended that the lift thickness be three times the nominal maximum aggregate size for fine graded mixes and four times the nominal maximum aggregate size for the coarse graded mixes. These same criteria will apply for SMA mixes. For OGFC mixes the thickness should be two times the nominal maximum aggregate size.

4.10.3 Determining the Appropriate Mix Type

The Expert Task Group that was formed to help write Information Series 128 provided the recommendations shown in Figure 4.40. The traffic level definitions of Table 4.12 along with the base, intermediate and surface layer definitions described earlier in this section are utilized in this figure. The appropriate mix types and sizes of mix (using nominal maximum aggregate size) are determined for each combination of traffic level and layer type. Also provided in Figure 4.40 is the recommended range of lift thickness for a given mix type and size. Additional guidance in utilizing this chart can be found in Information Series 128.

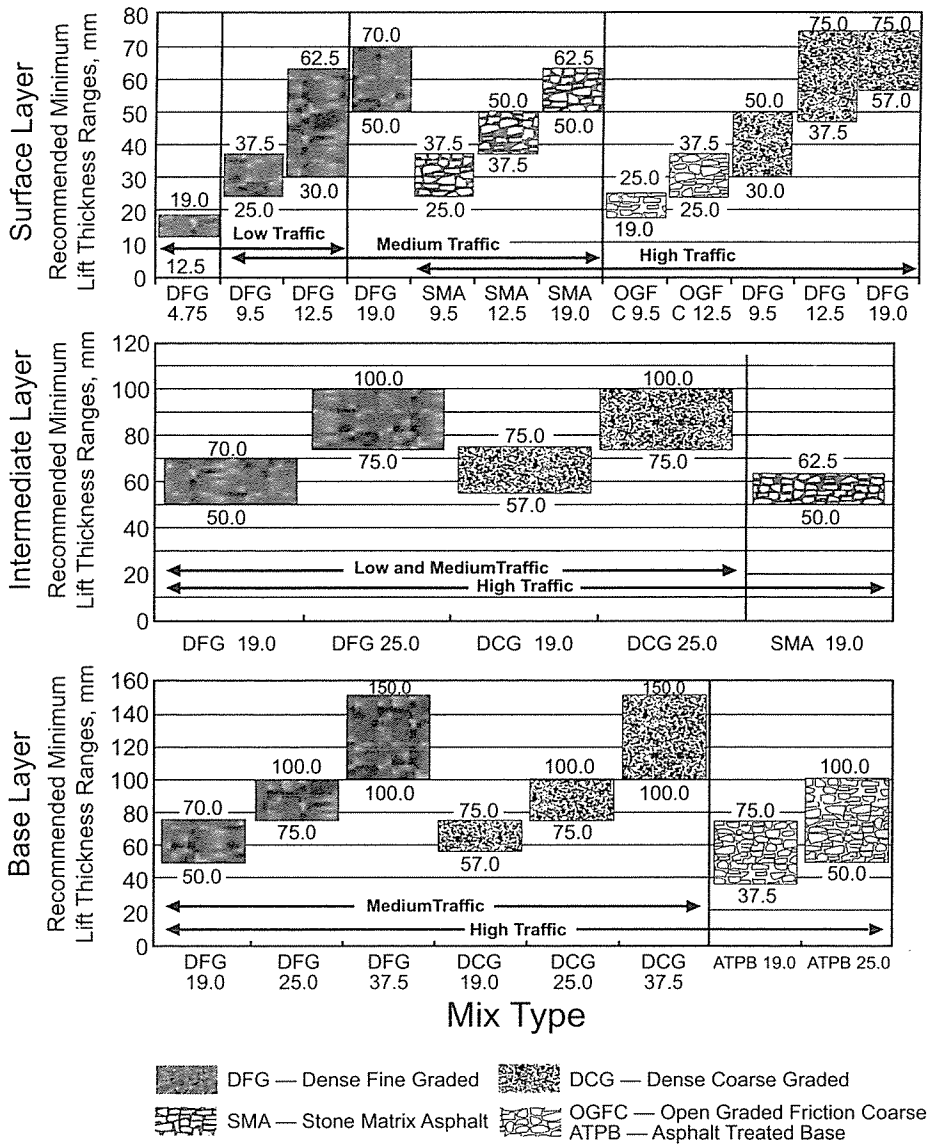


Figure 4.40 Recommended Mix Types

Resources

1. AASHTO Standard Practice for *Superpave Volumetric Design for Hot Mix Asphalt (HMA),* ASSHTO R 35.
2. AASHTO Standard Specification for *Designing Stone Matrix Asphalt (SMA),* AASHTO MP 08.
3. *Bailey Method for Gradation Selection in Hot Mix Asphalt Mixture Design.* Transportation Research Circular E-C044. Transportation Research Board, Washington, DC. October 2002.
4. Brown, E. R., and Mallick, Rajib B. *Stone Matrix Asphalt Properties Related to Mixture Design.* National Center for Asphalt Technology Report 94–02. Auburn, AL 1994.

5. *Designing and Controlling SMA Mixtures—State-of-the-Practice.* National Asphalt Pavement Association QIP-122. Lanham, MD 1999.
6. *Design, Construction and Maintenance of Open-Grade Asphalt Friction Courses.* National Asphalt Pavement Association, Information Series 115.
7. *FHWA Technical Advisory Open Graded Friction Courses, T5040.31.* Federal Highway Administration. December 1990.
8. Harvey, John T., Tsai, Bor-Wen Long, Fenella Hung, and Dave. *CALAPT Program—Asphalt Treated Permeable Base (ATPB).* California Department of Transportation Report FHWA/CA/OR-99/09. July 1999.
9. Kandhal, Prithvi S., and Mallick, Rajib B. *Design of New-Generation Open-Graded Friction Courses,* National Center for Asphalt Technology Report 99–03. Auburn, AL 1999.
10. Monismith, Carl L. "State of The Art: Bituminous Materials Mix Design." *Asphalt Mix Design and Construction—Past, Present and Future.* American Society of Civil Engineers. Reston, Virginia 2006.
11. *Recommended Use of Reclaimed Asphalt Pavement in the Superpave Mix Design Method: Guidelines.* National Cooperative Highway Research Program Research Results Digest No. 253. Transportation Research Board. Washington, DC.
12. Roberts, Freddy L, Kandhal, Prithvi S., Brown, E. Ray, Lee, Dah-Yin, and Kennedy, Thomas. *Hot Mix Asphalt Materials, Mixture Design and Construction.* 2nd ed. NAPA Education Foundation. Lanham, MD.
13. Vavrik, W. R., Pine, W. J., Huber, G.A., Carpenter, S.H., and Bailey, R. "The Bailey Method for Gradation Evaluation: The Influence of Aggregate Gradation and Packing Characteristics on Voids in Mineral Aggregate." *Journal of the Association of Asphalt Paving Technologists.* Vol. 70. 2001.

Suggested Reading

- *The Asphalt Handbook.* MS-4. The Asphalt Institute. Lexington, KY.
- *HMA Construction.* MS-22. The Asphalt Institute. Lexington, KY.
- *Mix Design Methods.* MS-2. The Asphalt Institute. Lexington, KY.
- *Superpave.* SP-2. The Asphalt Institute. Lexington, KY.

CHAPTER 5

SOILS AND DRAINAGE FOR ASPHALT PAVEMENTS

This chapter is an overview of subgrade classification methods and the tests used for evaluating foundation soils for highway, airport, and other pavement engineering purposes. Knowledge of these classification methods and strength evaluation tests are invaluable to the design and construction of asphalt pavements.

Section 5.1 Soil-Subgrade Classification Systems
- 5.1.1 Introduction
- 5.1.2 Objectives for Classifying Subgrade
- 5.1.3 Classification Systems

Section 5.2 Methods for Evaluating Subgrade Strength and Stiffness
- 5.2.1 Introduction
- 5.2.2 Laboratory Test Methods
- 5.2.3 Field Test Procedures
- 5.2.4 Strength Correlations

Section 5.3 Drainage of Asphalt Pavement Structures
- 5.3.1 Introduction
- 5.3.2 Surface Drainage
- 5.3.3 Design for Large Paved Areas
- 5.3.4 Subsurface Drainage
- 5.3.5 Geosynthetic Fabrics

5.1 Soil-Subgrade Classification Systems

5.1.1 Introduction

The performance of an HMA pavement is directly related to understanding the soils upon which it is built. The strength of both the untreated base course and the subgrade soil are directly related to the moisture content; therefore, the drainage of the pavement is extremely important.

Likewise, drainage is also an important aspect in asphalt pavement design and construction. The latter part of this chapter reviews some key issues of pavement-related drainage.

If not thoroughly knowledgeable about soil mechanics and the effect of different soils and their drainage, the reader is encouraged to consult a qualified geotechnical engineer for further guidance.

5.1.2 Objectives for Classifying Subgrade

The objective behind using any soil classification system for road and airport construction is the ability to predict performance of a given soil subgrade on the basis of a few simple tests. These test results, along with the soil classification procedure and criteria, allow an engineer to identify the soil type and characterize it according to other soils with similar characteristics and properties.

Grouping soils into a classification system is not the final objective, but it should be regarded as a tool to understand soil action and behavior as they relate to subgrade performance.

5.1.3 Classification Systems

The two classification systems commonly used to describe soils used for pavement construction are the American Association of State Highway and Transportation Officials (AASHTO) Classification System for Soils and the Unified Soil Classification System (USCS). Both systems are described in detail in many soil mechanics textbooks. The reader is referred to *An Introduction to Geotechnical Engineering*, by R. D. Holtz and W. D. Kovacs.

The principal soil classification system used in the United States by highway engineers is the AASHTO system. This system was originally developed by the U.S. Bureau of Public Roads in about 1928. The present AASHTO system, which dates from 1945, was the result of an extensive revision by the Highway Research Board.

Under the AASHTO system, soils having approximately the same general load-carrying capacity and service characteristics are grouped together to form one of seven basic groups (A-1, A-2, A-3, A-4, A-5, A-6, and A-7). In general, the best highway subgrade soils are classified as A-1. Poorer soils are rated in numerical order, with the A-7 group being the poorest. The exception is the A-3 group. Soils classified A-3 are better subgrades than soils in the A-2 group. From this, it can generally be assumed for a given set of conditions that the thickness of the pavement section must progressively increase as the soil classification group increases from A-1 to A-7, except as noted.

The USCS is based on the Airfield Classification System developed by Professor Arthur Casagrande of Harvard University during World War II. In 1952, Casagrande, together with the Bureau of Reclamation and the U.S. Army Corps of Engineers, modified the system for a broad range of applications, including dams and foundations.

The USCS groups soils based on grain size and plasticity. The system is rooted in the concept that coarse-grained soils can be classified according to grain-size distribution, while it is more appropriate to classify fine-grained soils based on plasticity characteristics. These characteristics are evaluated by plotting the plasticity index (PI) versus the liquid limit (LL) on a standard plasticity chart. The position of the plotted points yields information for predicting soil behavior as an engineering construction material. The following properties form the basis of soil identification:

- Percentages of gravel, sand, and fines passing the No. 200 (75 microns)
- Shape of the grain-size distribution curve (well-graded or poorly-graded
- Plasticity and compressibility characteristics

A soil is given a descriptive name and letter symbol to indicate its principal characteristics.

5.2 Methods for Evaluating Subgrade Strength and Stiffness

5.2.1 Introduction

Thickness design of asphalt pavements depends to a large extent on the strength and/or stiffness of the finished subgrade. Common empirical methods of thickness design depend on a subgrade strength index such as the CBR or R-value. Mechanistic-empirical methods use the subgrade resilient modulus, a measure of material stiffness, to establish pavement thickness. In either method, characterization of the subgrade strength or stiffness is a fundamental part of thickness design (for further information, see Chapter 6, "Thickness Design").

The concept of testing soils at their anticipated moisture and density should be the governing criterion for thickness design.

Subgrade strength or stiffness of a soil or aggregate base or subbase depends on soil type and is closely related to density and moisture content. Variable strength or stiffness can be expected, depending on the degree of compaction and moisture content after the pavement has been completed. If experience has indicated that a certain soil cannot be compacted economically to high density, it would be unwise to perform design tests with the soil at a high-density value.

Design strength or stiffness should be based on values obtained from soil samples in their worst anticipated condition. To obtain such values generally involves testing a sample that has been soaked in water.

Design methods used by different agencies vary widely, and their approach to evaluating the strength or modulus of the unbound materials for design purposes also varies. Evaluation methods discussed here are probably the most widely used methods for asphalt pavement design for airfields and highways.

Both field and laboratory test methods have been established by various agencies over the years. Each has advantages and disadvantages that must be considered. Field-test methods have the advantage of testing the unbound layers (base and subgrade) in situ at the confining pressure and moisture content that exist at the site. This may also be a disadvantage in that the existing conditions may not represent the intended conditions after construction. For example, if a pavement is to be built over a saturated soil, a field test would provide a strength index for the soil in a saturated condition; however, drainage improvements to the site will likely change the subgrade moisture content after construction.

Laboratory tests may be advantageous because the unbound layers are tested under a wide range of confining pressure and moisture conditions,

depending on the test used. On the other hand, it is very difficult to replicate in situ conditions in a laboratory environment. Furthermore, the act of soil sampling alone has the effect of disturbing the soil no matter how carefully it is done. In summary, the method of evaluation should be carefully considered.

Table 5.1 presents the commonly used test methods for soil strength and/or stiffness.

5.2.2 Laboratory Test Methods

The CBR, with its numerous variations, is probably the most widely used strength index for designing asphalt pavement structures. This method was first developed by the California Division of Highways in the 1920s. It was modified by the U.S. Army Corps of Engineers for use on airfields during World War II and is still widely used in airfield pavement design and for highway and roadway design outside the United States. While primarily a laboratory test, field CBR test methods have been developed for use in evaluating soils in situ.

The Resistance Value (R-Value) test method for evaluating treated and untreated materials for bases, subbases, and subgrades was developed by F. N. Hveem and R. M. Carmany of the California Division of Highways.

The Resilient Modulus (M_r) method was developed by the Asphalt Institute based on a test originated at the University of California by H. B. Seed, C. K. Chan, and C. E. Lee. This laboratory test is used to determine the stiffness of a soil specimen for conditions that represent stress in pavements subjected to moving wheel loads.

California Bearing Ratio

General

The U.S. Army Corps of Engineers test procedure was generally used until 1961 when the American Society for Testing and Materials adopted the method as ASTM D 1883, *Bearing Ratio of Laboratory-Compacted Soils*. The ASTM procedure differs in some respects from the Corps' procedure and the AASHTO procedure, AASHTO T 193. The ASTM procedure is the easiest to use and is the version described in this section.

Table 5.1
Common Test Methods for Soil Strength/Stiffness

Test Objective	Laboratory Test Methods	Field Test Methods
Strength Index	California Bearing Ratio (CBR) method	California Bearing Ratio (CBR) method
	Resistance Value Method (R-Value)	Dynamic Cone Penetrometer (DCP)
Stiffness	Resilient Modulus (M_r) method	Falling Weight Deflectometer (FWD)

The CBR is a comparative measure of the shearing resistance of soil. The test consists of measuring the load required to cause a piston of standard size to penetrate a soil specimen at a specified rate. This load is divided by the load required to force the piston to the same depth in a standard sample of crushed stone. The result, multiplied by 100, is the value of the California CBR. Usually depths of 0.1 or 0.2 inch (2.5 or 5 millimeters) are used, but depths of 0.3, 0.4, and 0.5 inch (7.5, 10, and 12.5 millimeters) may be used if desired. Penetration loads for the crushed stone have been standardized. This test method is intended to provide the relative bearing value, or CBR, of base, subbase, and subgrade materials. Procedures are given for laboratory-compacted swelling, nonswelling, and granular materials. These tests are usually performed to obtain information that will be used for design purposes.

The CBR value for a soil will depend upon its density, molding moisture content, and moisture content after soaking. Because the product of laboratory compaction should closely represent the results of field compaction, the first two of these variables must be carefully controlled during the preparation of laboratory samples for testing. Unless it can be ascertained that the soil being tested will not accumulate moisture and be affected by it in the field after construction, the CBR tests should be performed on soaked samples.

Certain routine soil tests should be performed before conducting the CBR test. These tests are shown in Table 5.2.

Test Procedure

If the sample is to be soaked, a representative sample of the material is taken to determine the moisture at the beginning of compaction, and another sample of the remaining material is taken after compaction. The material is weighed immediately and dried in an oven at 230±9°F (110±5°C) for at least 12 hours, or to constant weight. Each moisture-content sample should weigh not less than 100 grams for fine-grained soils or less than 500 grams for granular soils. If the sample is not to be

Table 5.2
Routine Soil Tests

	ASTM	AASHTO
Sieve Analysis of Fine and Coarse Aggregate	C 136	T 27
Liquid Limit of Soils	D 4318	T 89
Plastic Limit and Plasticity Index of Soils	D 4318	T 90
Particle Size Analysis of Soils (only for classification purposes)	D 422	T 88
Moisture-Density Relations of Soils Using 5.5-pound (2.5 kilograms) Rammer	D 698	T 99
Moisture-Density Relations of Soils Using 10-pound (4.5 kilograms) Rammer	D 1577	T 180

soaked, a representative sample of material is taken from one of the cut faces after penetration to determine the moisture content.

The apparatus for the CBR test is shown in Figure 5.1. The mold, with extension collar attached, is clamped to the base plate. The spacer disk is inserted over the base plate, and coarse filter paper is placed on top of the spacer disk. The soil sample is compacted at maximum density and optimum moisture content as determined by ASTM D 698 or D 1557.

The extension collar is removed, and the compacted soil trimmed even with the top of the mold with a straightedge. The spacer disk is removed and the mold inverted. The trimmed edge of the soil is placed on the perforated base plate, which has been covered with filter paper. The adjustable stem and plate assembly is placed on top of the compacted soil specimen in the mold.

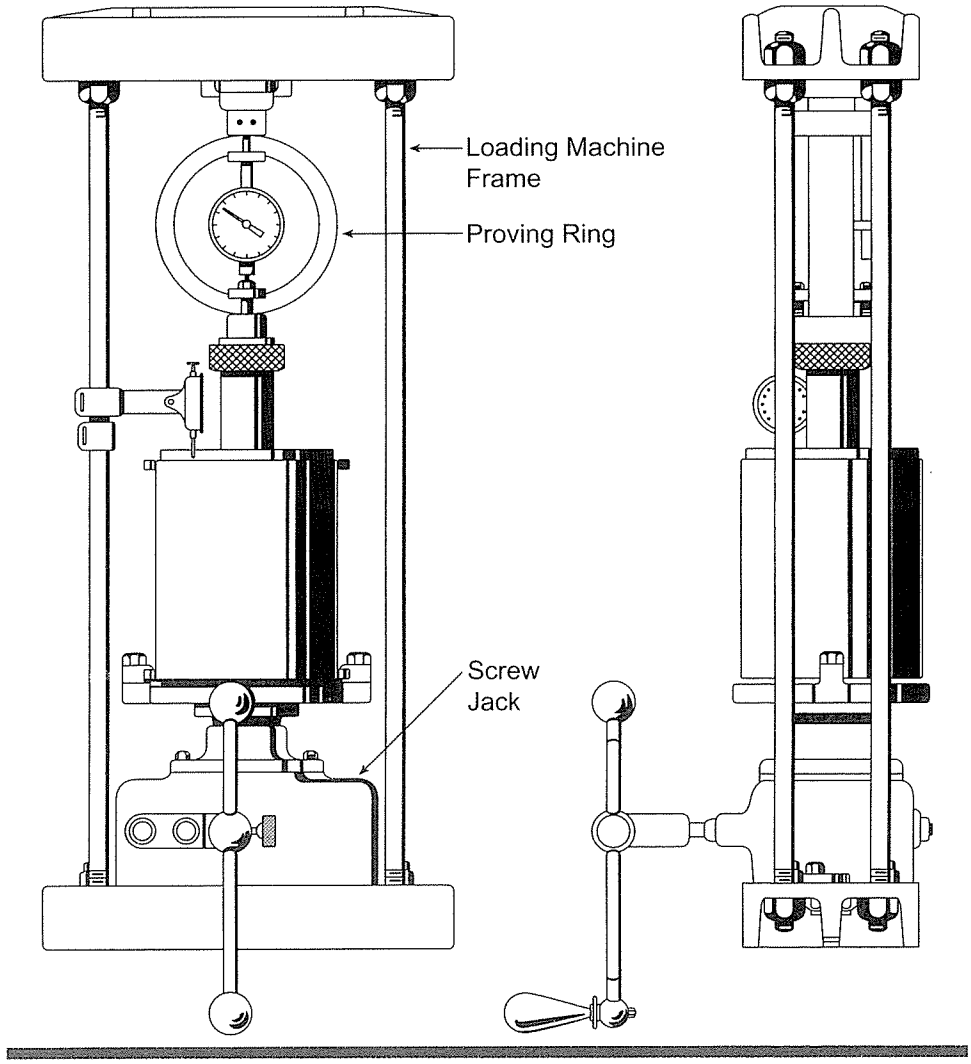


Figure 5.1 CBR Test Assembly

Surcharge weights within 5 pounds (2.27 kilograms) of the weight of the base material and pavement are added. In no case, however, should the surcharge be less than 10 pounds (4.54 kilograms). The sample is immersed in water and initial measurements taken for swell. After 96 hours, or a shorter period for soils that readily take up moisture, final swell measurements are made to determine swell as a percentage of initial specimen height. The sample is removed from the water and the surcharge weights and perforated plate are removed. The sample is allowed to drain for 15 minutes and then weighed and prepared for the penetration test.

The penetration test is conducted with surcharge weights in place, equal to those used during the soaking period. The mold assembly is placed in a loading frame or hydraulic press. The piston is first seated with an initial 10-pound (44 Newton) load, and all dials are set at zero. A load is then applied at the uniform rate of 0.05 inch (1.27 millimeters) of penetration per minute. The total load readings are taken at the following penetrations shown in Table 5.3:

Table 5.3
Measurement Points for CBR Test

0.025 in (0.6 mm)	0.125 in (3.2 mm)	0.250 in (6.4 mm)
0.050 in (1.3 mm)	0.150 in (3.8 mm)	0.300 in (7.6 mm)
0.075 in (1.9 mm)	0.175 in (4.4 mm)	0.400 in (10.1 mm)
0.100 in (2.5 mm)	0.200 in (5.1 mm)	0.500 in (12.7 mm)

Finally the load is released, the mold assembly disassembled, and the moisture-content is determined for the top 1 inch (25 millimeters), or for the entire depth of the sample.

Presentation of Data

After the test has been completed, the penetration load in pounds per square inch (megapascals) is calculated and the load-penetration curve plotted. In order to obtain true penetration loads from the test data, the zero point of the curve is adjusted to correct for surface irregularities and the initial concave upward shape of the curve, if present. If the curve is uniform as in example No. 1 of Figure 5.2, the CBR value is calculated from the recorded loads. For surface irregularities, as in example No. 3 of Figure 5.2, extend the straight-line portion of the curve to the base to obtain a corrected origin, or zero. If the curve has a reverse bend, or concave upward shape, as in example No. 2, draw a line tangent to the steepest point of the curve (A), and extend the line to the base to obtain a corrected origin or zero point (B). Then read the corrected load values for 0.1 inch (2.5 millimeters) penetration (C) and 0.2 inch (5 millimeters) of penetration (D).

The CBR value is defined as a ratio comparing the bearing of a material with the bearing of a well-graded crushed stone. The penetration loads for crushed stone are presented in Table 5.4.

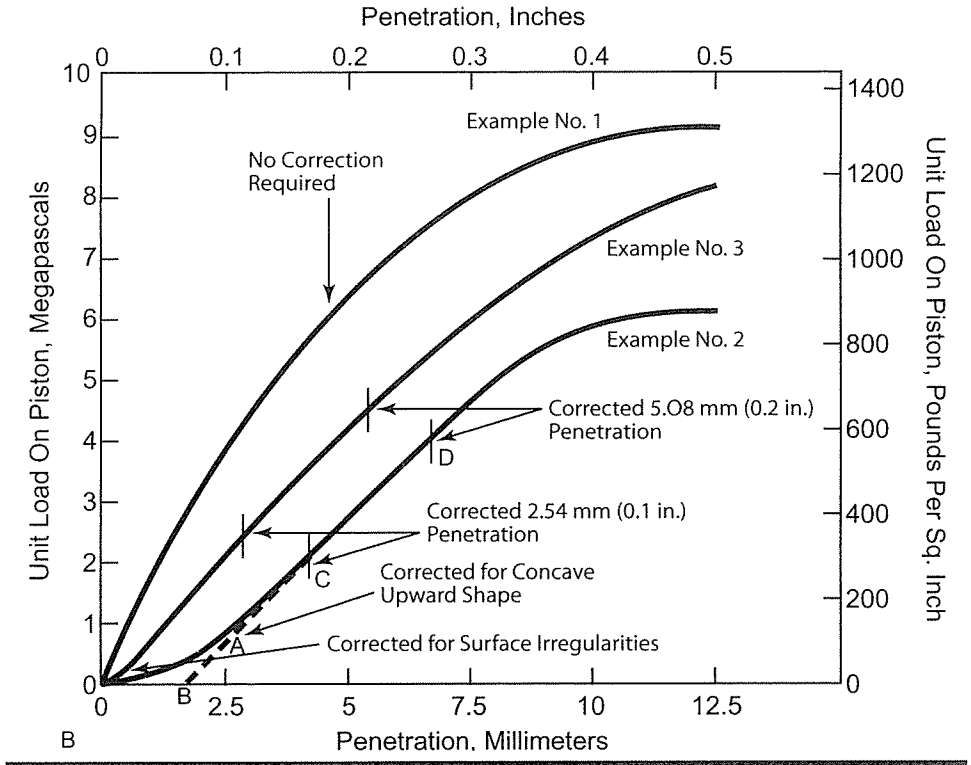


Figure 5.2 Correction of Load-Penetration Curves

Table 5.4
Standard Penetration Loads for CBR Test

Penetration		Standard Load		Standard* Load	
inch	(mm)	lb	(N)	psi	(MPa)
0.1	(2.5)	3,000	(13,345)	1000	(6.89)
0.2	(5.0)	4,500	(20,017)	1500	(10.34)
0.3	(7.5)	5,700	(25,355)	1900	(13.10)
0.4	(10.0)	6,900	(30,693)	2300	(15.86)
0.5	(12.5)	7,800	(34,696)	2600	(17.93)

*Plunger cross-section area = 3 in^2 (1,935.5 mm^2)

The CBR is determined from the corrected load values at 0.1- and 0.2-inch (2.5 and 5 millimeters) penetrations by dividing the loads at 0.1 and 0.2 inch (2.5 and 5 millimeters) by the standard loads of 1000 and 1500 pounds per square inch (6.89 and 10.34 megapascals), respectively. Each ratio is multiplied by 100 to obtain the CBR in percentage. The CBR is usually selected at 0.1 inch (2.5 millimeters) of penetration. If the CBR at 0.2 inch (5 millimeters) of penetration is greater, the test should be rerun. If check tests give similar results, the CBR at 0.2 inch (5 millimeters) of penetration should be used. Figure 5.3 is a suggested form for recording CBR data.

CALIFORNIA BEARING RATIO TEST DATA SHEET

Molding Date _____ Penetration Date _____

Project _____

Sample No. _____ Compacted at _____

A. Weight of Compacted Sample, Mold and Base Plate; kg _____
B. Tare Weight of Mold, and Base Plate; kg _____
C. Weight of Sample; kg _____
D. Height of Compacted Sample; mm _____
E. Volume of Sample; cm^3 _____
F. Unit Wet Weight; kg/cm^3 _____
G. Moisture Content; % _____
H. Unit Dry Weight; kg/cm^3 _____

Expansion and Consolidation Data

N. Surcharge Weight _____ kg R. % Expansion or Cons. _____
O. Dial Reading at Start _____ mm
P. Dial Reading at Finish _____ mm
Q. Difference _____ mm

$$R = \frac{Q}{D} \times 100$$

S. Weight of Sample, Mold and Base Plate after Saturation; kg _____

Moisture Samples

Top 25 mm				Bulk of Sample			
Pan No.				Pan No.			
Wet Wt.	g	Dry Wt.	g	Wet Wt.	g	Dry Wt.	g
Dry Wt.	g	Tare Wt.	g	Dry Wt.	g	Tare Wt.	g
Percent Moisture				Percent Moisture			

T. Sum of Net Wet Weights _____ + _____ = _____
U. Sum of Net Dry Weights _____ + _____ = _____

Penetration Data

Pen.	Load, kN (lb)	Load, MPa (psi)
0.6 mm (.025 in.)	:	
1.3 mm (.050 in.)	:	
1.9 mm (.075 in.)	:	
2.5 mm (.100 in.)	:	
3.2 mm (0.125 in.)	:	
3.8 mm (.150 in.)	:	
4.4 mm (0.175 in.)	:	
5. mm (.200 in.)	:	
6.4 mm (0.250 in.)	:	
7.5 mm (.300 in.)	:	
10. mm (.400 in.)	:	
12.5 mm (.500 in.)	:	

Corrected C.B.R.

Pen.	Corr. Load MPa (psi)	Std. MPa (psi)	C.B.R.
2.5 mm (.100 in.)		6.89 (1000)	
5. mm (.200 in.)		10.34 (1500)	
7.5 mm (.300 in.)		13.10 (1900)	
10. mm (.400 in.)		15.86 (2300)	
12.5 mm (.500 in.)		17.93 (2600)	

NOTE: $C = A - B$ $F = \frac{C}{E}$

$$G = \frac{C-U}{U} \times 100 \qquad H = \frac{F}{100 + G} \times 100$$

Figure 5.3 Sample Form for Recording CBR Data

Resistance Value (R-Value)

General

The R-value method of design is based on two separate measurements: the R-value (or resistance value) and the expansion pressure test.

The cover thickness or structural section required to prevent plastic deformation of soil under imposed wheel loads is determined by the R-value. An expansion pressure test is used to determine the thickness or

weight of cover required to maintain soil compaction. The design R-value is determined from the moisture content and density at which these two thicknesses are equal. With granular nonexpansive soils, the design R-value is determined for a density considered to be equivalent to that obtained by normal construction compaction. This density value is obtained from the exudation pressure data.

The test method for determining the R-value consists of the following steps:

1. Preparation of samples
2. Compaction of test specimens
3. Determination of exudation pressure of test specimens
4. Determination of expansion pressures of test specimens
5. Determination of R-value

Test Procedure

The R-value test (ASTM D 2844) is made on a relatively small sample, and it is necessary to remove all large material. If 75 percent or more of the material passes the ¾-inch (19.0 millimeters) sieve, the material retained on that sieve is removed. If less than 75 percent of the material passes the ¾-inch (19.0 millimeters) sieve, then all material retained on the 1-inch (25.0 millimeters) sieve is removed. The soil is then separated over the No. 4 (4.75 millimeters) sieve but recombined in the proper proportions with the coarse sizes to obtain 1200-gram (2.65 pounds) samples plus an extra 200 grams (0.44 pounds) for moisture content determinations.

In each R-value test, four test briquettes at different moisture contents are prepared. Each briquette has one-half to two-thirds the amount of water necessary to saturate the sample or exude moisture between 100 and 800 pounds per square inch (689 and 5516 kiloPascals). After mixing, loose material is allowed to cure overnight in a covered container. Then more water is added and mixed until the estimated saturation moisture is attained. The first briquette is used as a pilot specimen to obtain specimen heights of 2.5 ± 0.1 inches (63.5 ± 2.5 millimeters) and to have exudation pressures both above and below 300 pounds per square inch (2068 kiloPascals).

Specimens are compacted by a kneading compactor, which is the same as that used for the Hveem method of asphalt mix design. The compaction procedure provides a kneading action intended to give soil particles the same orientation and contact pressure obtained during field compaction.

The prepared soil sample is placed in the compactor feeding trough with the material distributed evenly along the full length. A 4-inch (101.6 millimeters) diameter by 5-inch (127 millimeters) tall mold is fitted into a special holder for placing in the compactor. Material is deposited in the mold by a special procedure while the kneading compactor is operating and the foot pressure is set at 250 pounds per square inch (1724 kiloPascals).

When all material has been placed in the mold and 10 extra blows have been added to level and seat the material, the sample is compacted by using 100 blows with a foot pressure of 350 pounds per square inch (2413 kiloPascals).

Some clays and clean sands require lower compaction pressures. It may be necessary, if the surface is uneven, to smooth and level the tamped surface.

The exudation pressure test is used to determine the compressive stress necessary to exude water from the compacted specimen. Results from tests on several specimens are used to determine the R-value at a standard pressure of 300 pounds per square inch (2070 kiloPascals).

In this test, a bronze disk is placed on top of the tamped specimen in the mold, and a piece of filter paper is placed on the disk. The assembly is then inverted so that the filter paper is on the bottom. Before the assembly is positioned in the testing machine, it is placed on the contact plate of the moisture exudation indicator (see Figure 5.4).

A ram is used to force the sample down in the mold to the contact plate of the exudation indicator. The testing machine then applies a load to the ram at the rate of 2000 pounds (8.896 kiloNewtons) per minute until water exuding from the sample on the indicator plate lights five of the six outer lights of the indicator. Total load on the sample at that moment is converted to pounds per square inch (kiloPascals) and recorded as exudation pressure. Specimens having exudation pressures of less than 100 pounds per square inch (689 kiloPascals) or more than 800 pounds per square inch (5520 kiloPascals) are usually discarded.

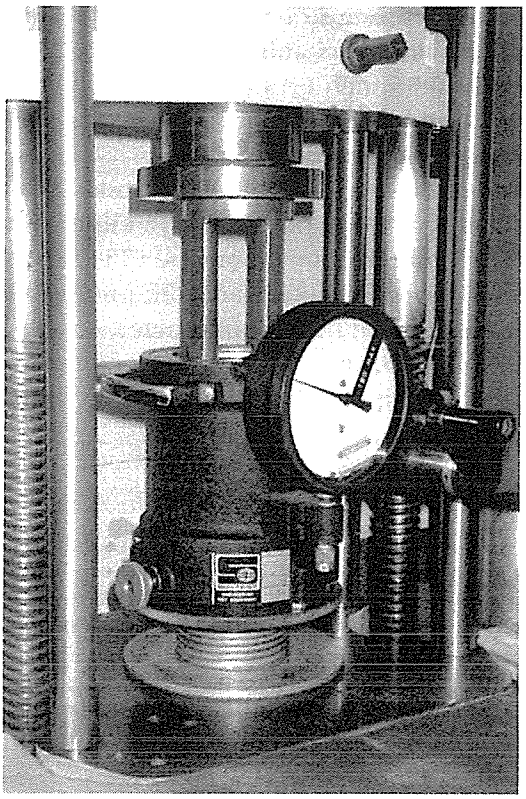


Figure 5.4 Hveem Stabilometer

After the test is completed, the specimen is left in a covered mold for at least a half hour before the next part of the test procedure is begun.

The expansion pressure test is used to determine the expansion pressure a soil develops under water. This test is performed in such a way that a large change in sample density does not occur during testing. Pavement thicknesses required to withstand expansion pressures can be determined and compared with those determined by the R-value test.

The specimen is allowed to stand for at least a half hour after the completion of the exudation test. A perforated brass plate with stem is placed on the face of the compacted specimen in the mold. The mold is then positioned in the calibrated expansion device, and 200 milliliters of water is poured on the specimen in the mold. Expansion pressure is allowed to develop in 16 to 24 hours, after which the deflection of the calibrated spring steel bar is read and the expansion pressure determined.

The R-value test provides a rapid means for determining a measure of the stability or resistance to plastic deformation of compacted materials by the Hveem Stabilometer (see Figure 5.4). It is the same device used for the Hveem method of asphalt mix design.

The specimen that was tested for expansion pressure is forced from the mold into an adjusted stabilometer. The stabilometer is then positioned in the testing machine. The head is lowered until it contacts the specimen but applies no load. A lateral pressure of 5 pounds per square inch (35 kiloPascals) is then applied with the stabilometer pump. Next, the specimen is loaded at the rate of 0.05 inch (1.3 millimeters) per minute and lateral pressure readings are obtained at 80 and 160 pounds per square inch (550 and 1100 kiloPascals) vertical pressure.

Vertical loading must be stopped at 2000 pounds (8.896 kiloNewtons) and reduced immediately to 1000 pounds (4.448 kiloNewtons). The lateral pressure is reduced to 5 pounds per square inch (35 kiloPascals). The turns displacement dial indicator is set at zero and the pump handle is turned at two revolutions per second until the stabilometer gauge reads 100 pounds per square inch (689 kiloPascals). The number of turns is recorded, and the stabilometer R-value is then calculated from the following formula:

$$R = \frac{100}{\frac{2.5}{D}\left(\frac{P_v}{P_h} - 1\right) + 1}$$

where

D = turns displacement reading
P_v = 160 pounds per square inch (1100 kiloPascals) vertical pressure
P_h = horizontal pressure [stabilometer gauge reading 160 pounds per square inch (1100 kiloPascals) vertical pressure]

Presentation of Data

Thickness design of asphalt pavements will be discussed in detail in Chapter 6, "Thickness Design." The design R-value, however, cannot be

determined without using a thickness design chart. The reader is referred to MS-1, *Thickness Design—Asphalt Pavements for Highways and Streets* for thickness design charts.

Resilient Modulus (M_r)

General Theory

The resilient modulus (M_r) is used to characterize subgrade soils and unbound base/subbase materials. The resilient modulus test provides a material stiffness that more closely simulates the behavior of the material under a moving wheel. It can be used in the mechanistic analysis of multilayered systems for predicting roughness, cracking, rutting, etc. This test covers the determination of resilient modulus of untreated fine-grained soils for conditions that represent stress in pavements subjected to moving wheel loads. The results of the test describe the resilient character of a soil specimen when a repeated axial deviator stress of fixed magnitude, frequency, and load duration is applied while the specimen, in a triaxial chamber, is surrounded by pressure that acts as a static, all-around stress.

Test Procedure

Resilient modulus testing of unbound materials is performed in accordance with one of two current procedures: (AASHTO T 307), *Determining the Resilient Modulus of Soils and Aggregate Materials* or (AASHTO T 292), *Resilient Modulus of Subgrade Soils and Untreated Base/Subbase Materials.* A triaxial cell suitable for use in resilience testing of soils is shown in Figure 5.5. The equipment is similar to most standard cells, except that it is somewhat larger to facilitate the internally mounted load and deformation measuring equipment, and has additional outlets for the electrical leads from the measuring devices. (Triaxial cells with suitable externally mounted load and deformation measuring equipment also may be used.) For the type of equipment shown, air is used as the cell fluid.

The deformation measuring equipment consists of Linear Variable Differential Transformers (LVDTs) attached to the soil specimen by a pair of clamps. Two LVDTs are used for the measurement of axial deformation. The clamps and LVDTs are shown in position on a soil specimen in Figure 5.5. Preferably, load is measured by placing a load cell between the sample cap and the loading piston. In addition to measuring devices, suitable signal excitation, conditioning, and recording equipment must also be maintained.

The external loading source may be any device capable of providing varying repeated loads in fixed cycles of load and release. These devices range from simple cam and switch control of static weights or air pistons to closed loop electro-hydraulic systems. A load duration of 0.1 second and load-release duration of 0.4 to 3 seconds has been found to be satisfactory for most applications.

The confining pressure and deviator stress used should approximate the expected in-place state of stresses. A confining pressure of 2 pounds per square inch (14 kiloNewtons) and deviator stress of 6 pounds per square inch (41 kiloNewtons) are reasonable values for most testing.

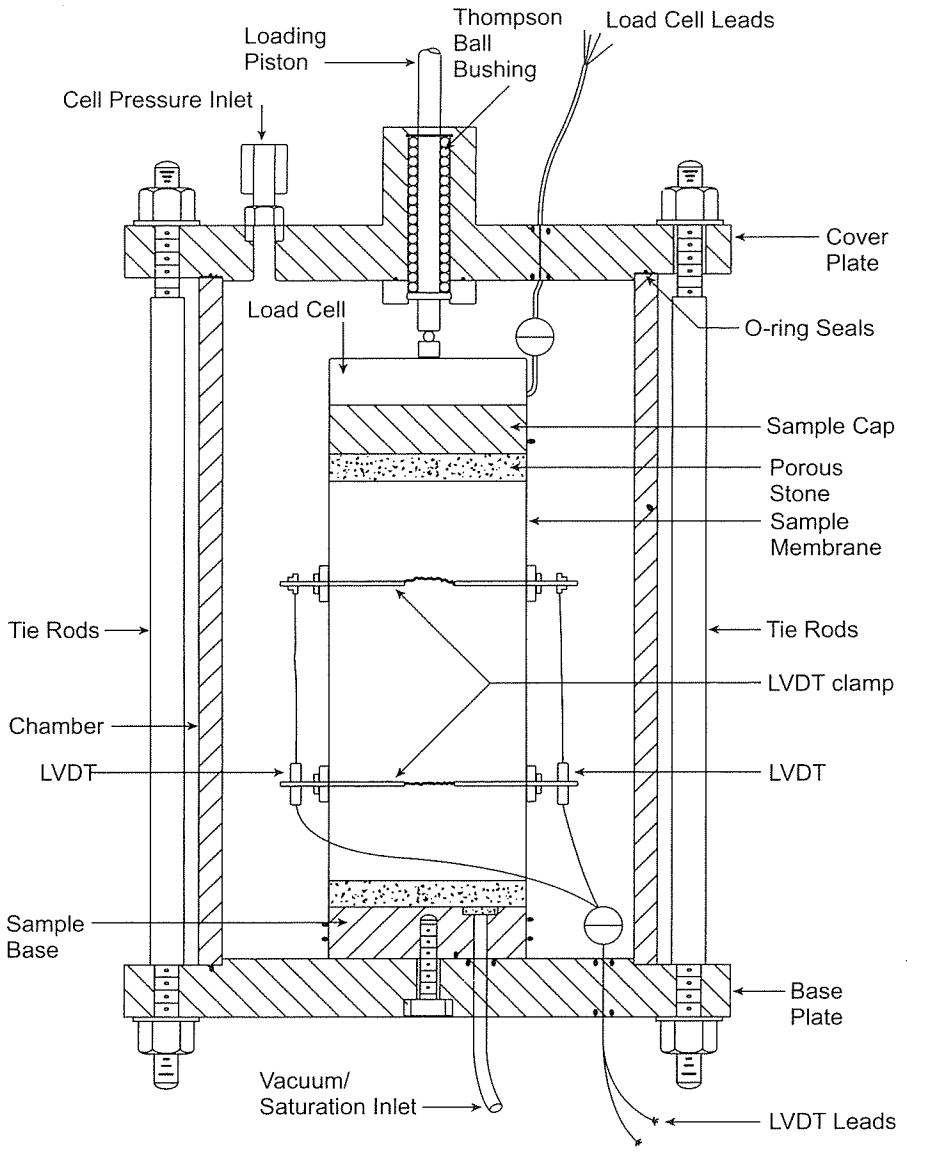


Figure 5.5 Apparatus for Resilient Modulus Testing of Soils

The test is started by applying 200 repetitions of a deviator stress of approximately 3 pounds per square inch (21 kiloPascals), followed successively by 200 repetitions each at 6 pounds per square inch and 9 pounds per square inch (41 kiloPascals and 62 kiloPascals). The foregoing stress sequence constitutes sample conditioning, that is, elimination of the effects of the interval between compaction and loading, and also the effects of initial load versus reloading. The deviator load is decreased to 6 pounds per square inch (41 kiloPascals), after which 200 repetitions of load are applied and deviator load and vertical recoverable deformation at or near the 200th repetition are recorded. At the completion of the load, the chamber pressure is reduced to zero.

Presentation of Data

Test data can be recorded and calculations performed using a table such as that illustrated in Figure 5.6. For specific procedures, refer to MS-10, *Soils Manual* from the Asphalt Institute.

5.2.3 Field Test Procedures

Dynamic Cone Penetrometer

Field Test Methods

The Dynamic Cone Penetrometer (DCP) is a field-testing device to quickly estimate the in situ strength and deformation characteristics of pavement subgrades. The test can be performed on native soils or on compacted materials. The DCP Index that results from completing the test can be readily correlated to a CBR for use in the empirical CBR design method. Developed in South Africa in 1975, the DCP has since found its application in many parts of the world because of its portability, speed, and ease of use in flexible pavement design.

For field evaluation of subgrade stiffness, the Falling Weight Deflectometer (FWD) is commonly used by highway agencies and airports throughout the United States. The test is conducted by raising and dropping a weighted package on the pavement surface and measuring vertical deflections on the pavement surface. Using principles of mechanics, the stiffness of the HMA layers, aggregate base, and subgrade layers can be back-calculated at the testing location. Due to the speed and nondestructive nature of the test, numerous tests can be conducted on a pavement feature in a relatively short amount of time.

General

The DCP is a testing device used to estimate the in situ strength and deformation characteristics of native soils or fills placed as part of a pavement subgrade; it generally is not useful to evaluate the properties of bases and subbases. If the test is performed on existing pavements, the asphalt concrete surface layer must be removed, typically by coring an appropriate-size hole in the asphalt concrete. It relies on correlations with various soil types and the relative compaction or density of soils. Though commonly used to estimate the structural capacity of soils, this test is now being considered as a quality-control tool for pavement subgrades during construction, in lieu of more common in situ density testing.

The test involves driving a cone-shaped probe (see Figure 5.7) into a soil using a dynamic load and measuring the advancement of the probe for each applied load. The strength of the soil being penetrated has a direct impact on the penetration observed. Thus, the strength of the soil is related to the depth of penetration achieved with each blow used to drive the DCP. As previously discussed, the strength of the soil depends on density, moisture content, and soil type or classification.

One of the limitations of the DCP is that it may be incapable of penetrating through granular base layers. In these cases, it may be best to

Figure 5.6 Forms for Resilient Modulus Tests on Compacted Fine-Grained Soils

Soil Sample _____
Location _____
Sample No. _____
Specific Gravity _____

Soil Specimen Measurements

Diameter
Top _____
Middle _____
Bottom _____
Average _____
Membrane Thickness _____
Net Diameter _____
Ht Specimen + Cap + Base _____
Ht Cap + Base _____
Initial Length L_0 _____

Soil Specimen Weight
Initial Wt of Container
+ Wet Soil-gms _____
Final Wt of Container
+ Wet Soil-gms _____
Wt Wet Soil Used _____

Soil Specimen Volume
Initial Area L_0
cm^3 ($in.^3$) _____
Volume A_0L_0
cm^3 ($in.^3$) _____
Wet Density-kg/m^3 (lb/ft^3) _____
Water Content-% _____
% Saturation _____
Dry Density-kg/m^3 (lb/ft^3) _____

Date _____
Compaction Method _____
Vertical Spacing Between
LVDT Clamps-mm (in.) _____
Chamber Pressure-kPa (psi) _____

Constants
Vertical LVDT _____
Load Cell _____
Comments _____

Specimen No.	Load Cell Chart Reading	Deviator Load kg (lb)	σ_d kPa (psi)	Recoverable Deformation LVDT Chart Reading	Recoverable Deformation mm (in.)	ϵ_r mm/mm (in./in.)	$M_r = \frac{\sigma_d}{\epsilon_r}$ kPa (psi)

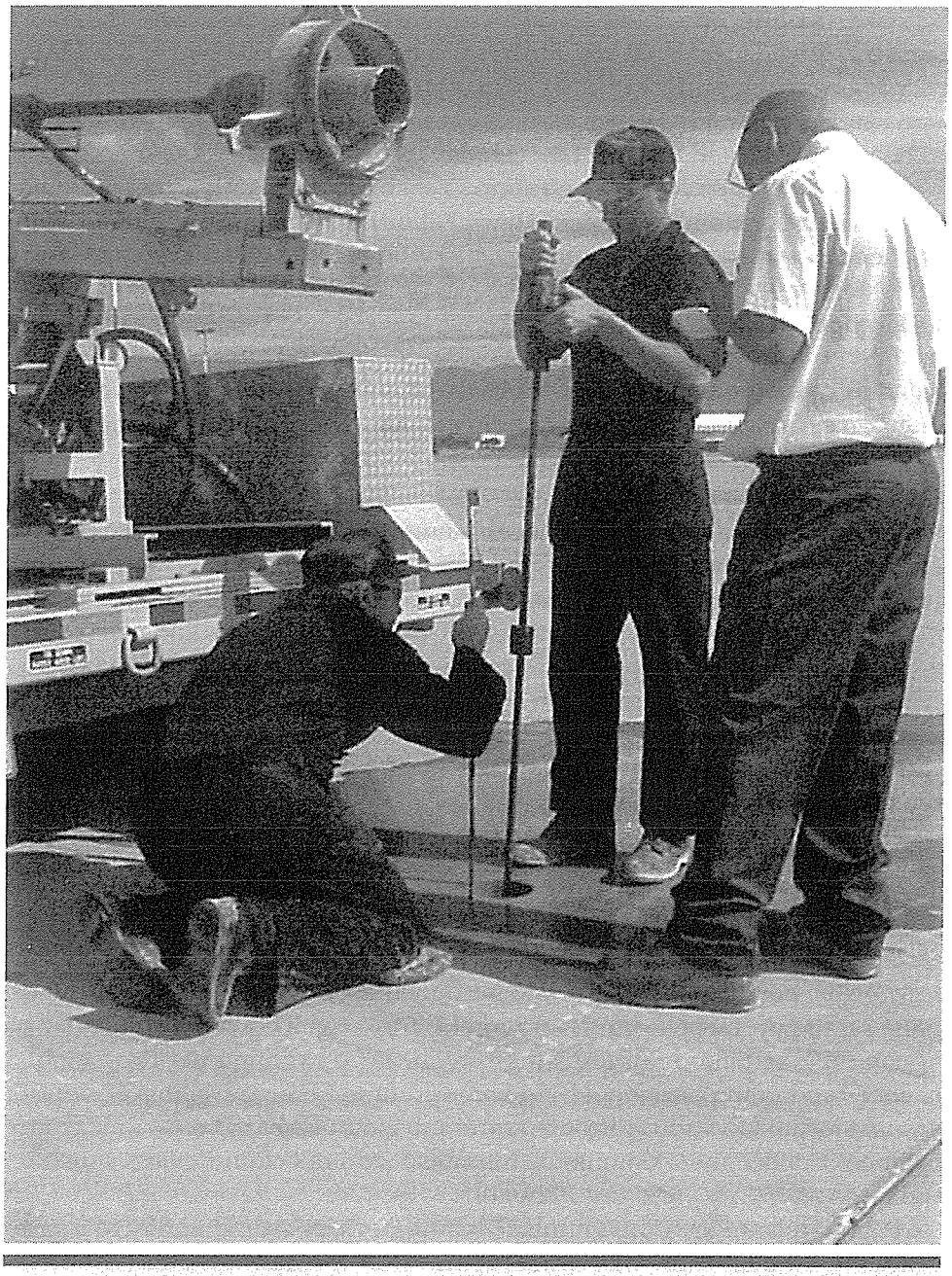


Figure 5.7 Hand-Operated DCP

auger through the granular base layer prior to attempting the DCP test on subgrade soil.

Test Procedure

The DCP consists of a 0.62-inch (15.8 millimeters) diameter steel rod with a cone-shaped tip, a hammer, a coupler assembly, and a handle. The test is conducted by dropping a 17.6-pound (8.0 kilograms) weight a fixed height of 22.6 inches (575 millimeters) and measuring the penetration of

the cone tip. As presented in ASTM 6951, the 60-degree cone tip has a diameter of 0.79 inch (20 millimeters). Generally, the device is made of stainless steel to protect it from corrosion, and the cone is made of hardened steel or a similar material.

The data recorded during the testing process include the number of blows and the depth of penetration. The rate of penetration is defined as the depth of penetration per blow, called the penetration index or the DCP index—expressed as millimeter/blow or inch/blow. The penetration rate is determined as the slope of the curve relating the number of blows to the penetration depth. The device can be operated manually or can be installed on a vehicle and automated. The automated DCP (see Figure 5.8) has a fully developed software tool to determine the penetration index. In either case, the equipment is economical, simple to use, and amenable to many types of evaluations and materials.

Presentation of Data

The data is presented as a plot of the number of blows versus the depth of penetration. The penetration rate has been correlated to engineering properties of soils or other derived measurements of strength and modulus. Specifically, studies have correlated CBR to the penetration rate for a wide variety of cohesive and granular materials. The penetration rate can be correlated to resilient modulus through correlations between CBR and resilient modulus; however, such secondary correlations should be used with caution. Several studies have also correlated penetration rate, or index, to the elastic modulus of soils.

Falling Weight Deflectometer

General

The FWD, as shown on Figure 5.9, is an impulse load-deflection device that delivers a transient impulse force to the pavement. A weight is lifted to a given height on a guide system and then dropped onto a buffering plate on the pavement. A transient impulse force, which can be changed by varying the magnitude of the falling weight or by varying the drop height, is generated in the pavement by the impact of the falling weight. A series of sensors located at various distances from the load plate are used to measure pavement deflections. Commercial impulse load-deflection devices include the Dynatest, KUAB, JILS, and Phoenix.

A major advantage of the impulse loading devices is their ability to more accurately model a moving wheel load in both magnitude and duration, thus producing a deflection that simulates the pavement deflection caused by a moving vehicle. The advantages of the FWD are the following:

- The ability to better simulate a moving wheel load
- The ability to measure deflections at varying loads over time
- The ability to record a deflection basin
- The speed, repeatability, and robustness of the equipment

Test Procedure

The FWD is moved to a test point, and the loading plate and transducers are hydraulically lowered to the pavement. A test sequence is then